Mohammad Vazir Panah

Embedded Systems Engineer · FPGA, RISC-V, C/C++

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LinkedIn · GitHub · Website

Summary

Embedded systems engineer with an M.Sc. in Computer Systems Architecture and three years of R&D experience on RISC-V and heterogeneous embedded platforms. I work across the hardware and software boundary: C/C++ on embedded CPUs and GPUs, FPGA design in VHDL and Verilog, board bring-up and toolchains. Used to reproducible engineering workflows (Git, Docker, CI, CMake) and to working in international project teams.

Technical Skills

Programming	C, C++, Python, Assembly
Embedded & Hardware	RISC-V, FPGA (Intel DE0-Nano, DE10-Nano-SoC), microcontroller and SoC architecture, FreeRTOS
Digital Design	VHDL, Verilog, Quartus, ModelSim, Avalon-MM, high-level synthesis (basic)
Hardware Test	Design for test: full-scan, ATPG, BIST, fault simulation
Toolchains & Debug	GCC, LLVM/Clang, RISC-V cross-compilation toolchains, low-level runtime debugging
Simulation & Modeling	SystemC, Gem5, QEMU, OVPsim, MATLAB/Simulink
Parallel & Heterogeneous	OpenMP, OpenCL, SYCL, CUDA
Tools	Git, Docker, CI, CMake, Linux
Certification	TinyML, Harvard University (edX)

Professional Experience

Research Fellow, HPC Group (ISISLab) *2023 - 2026*
University of Salerno, Department of Computer Science, Salerno, Italy

- Set up RISC-V toolchains, firmware and full embedded software stacks on physical boards in the EuroHPC LIGATE project, shared across several partner institutions
- Brought up and evaluated RISC-V boards (AllWinner D1, BananaPi-F3) and embedded SoCs for benchmarking and real-time workloads
- Developed and deployed portable real-time applications in C/C++ on heterogeneous embedded CPU/GPU platforms (SYCL, OpenMP, OpenCL) in the LibreRT project
- Found and fixed kernel dispatch failures on Imagination PowerVR embedded GPUs by tracing the issue into the LLVM-based AdaptiveCPP runtime; resolved related OpenCL build problems
- Built reproducible build and test pipelines (Git, Docker, CI, CMake) for RISC-V and SoC boards

Teaching Assistant *2020 - 2021*
Shahid Beheshti University, Tehran, Iran

- System-on-Chip, Fundamentals of Embedded & Real-Time Systems, Design & Analysis of Real-Time Embedded Systems, Advanced Computer Architecture
- Ran student FPGA and embedded lab projects from environment setup to lab sessions, feedback and grading

C++ Instructor and Hardware Intern *2013 - 2015*
Sepehr Educational Institutions and Abooreyhan Company, Kashmar, Iran

- Taught C++ programming courses; diagnosed and repaired PC hardware, assembled and configured systems

Selected Projects

Real-Time Tiny YOLO Inference and FIR Filter IP Core on FPGA *2019*
Intel DE10-Nano-SoC and DE0-Nano, GitHub

- Implemented a fixed-point real-time Tiny YOLO v1 object-detection pipeline on FPGA under tight resource limits
- Designed a FIR filter IP core in Quartus with an Avalon-MM interface

Custom CPU with Full-Scan DFT, BIST and LSSD Self-Test *2020*
Verilog, GitHub

- Designed a CPU in Verilog and applied scan insertion, fault collapsing and ATPG, reaching 100% fault coverage after scan insertion

- Built LFSR/MISR-based BIST and LSSD boundary-scan self-test for the same design

Parallel Sorting and Matrix Multiplication Hardware *2020*
Verilog and CUDA, GitHub

- Designed mesh- and hypercube-connected sorting networks and a shared-memory matrix multiplication architecture in Verilog

SADF Model of Computation in ForSyDe-SystemC (M.Sc. Thesis) *2022*
C++ and SystemC, GitHub

- Implemented the Scenario-Aware Dataflow model in C++ templates; the core process constructors were merged into the official ForSyDe-SystemC repository
- Connected the models to ROS and Gazebo to drive a simulated mobile robot in real time

Real-Time Audio Controller *2020*
UPPAAL and Lingua Franca, GitHub

- Verified timing properties with timed automata and re-implemented the controller as a reactive program, fixing clock drift between distributed timers

Education

M.Sc. Computer Engineering, Computer Systems Architecture *2019 - 2022*
Shahid Beheshti University, Tehran, Iran

- GPA 18.43/20 · Thesis: *A Framework for Modeling Self-Aware Cyber-Physical Systems with SystemC*

B.Sc. Computer Engineering, Hardware Engineering *2012 - 2016*
Birjand University, Birjand, Iran

- GPA 16/20 · Top 15% of Computer Engineering graduates

Publications

- B. Cosenza et al. (incl. **M. Vazir Panah**). *LibreRT: Portable Heterogeneous Real-Time Programming for the Embedded Computing Continuum.* ACM Computing Frontiers 2026.
- L. Carpentieri, **M. Vazir Panah**, B. Cosenza. *A Performance Analysis of Autovectorization on RVV RISC-V Boards.* IEEE PDP 2025.
- S.-H. Attarzadeh-Niaki, **M. Vazir Panah**. *System-Level Modeling of Dynamic Applications with Scenario-Aware Dataflow Graphs.* JICSE 2023.
- **M. Vazir Panah**, S.-H. Attarzadeh-Niaki, A. Salimi-Badr. *ROS-Based Co-Simulation for Formal Cyber-Physical Robotic System Design.* IEEE CSICC 2022.

Awards

- National R&D Research Grant, Mobile Telecommunication Company of Iran (2020), awarded in competition with Master's and PhD students

Languages

Persian (native) · English (C1) · Italian (B1 listening, A2 speaking and writing)

References available on request.